

CYREL JANE A. EDAÑO

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EDUCATION

University of San Carlos

Bachelor of Science in Computer Science

PROJECT EXPERIENCE

Predictive Energy Management via Topology Mapping

AI Full Stack Engineer

- Engineered a 4-stage physics-informed ML pipeline (data engineering → physics simulation → ML optimization → analysis output) combining XGBoost/LightGBM and a PPO RL agent to optimize MGU-K deployment across 4 F1 circuits, operating within strict FIA constraints of 120 kW power cap and 2 MJ/lap energy budget.
- Delivered circuit strategy reports across 4 circuits (Monza, Spa, Suzuka, Silverstone) with deployment heatmaps, SoC trace visualizations, and sensitivity ablations; fully containerized via Docker with a 6-command CLI pipeline orchestrator

F1 Telemetry Speed Analysis Dashboard

Data Analyst

- Leveraged the FastF1 Python library to process over 10,000+ telemetry data points per race, analyzing speed, throttle, and brake traces to identify lap time deltas with 0.01s precision.
- Developed interactive visualizations using Matplotlib and Seaborn, allowing for comparative analysis of driver performance that reduced post-race strategy review time by 50%.

Digital Burnout Early-Warning System

Machine Learning Engineer

- Engineered 30+ behavioral and linguistic features across 7-day and 30-day rolling windows, including 12 LIWC-style psycholinguistic categories and a 540-term MBI-aligned burnout lexicon spanning 4 subscale categories (emotional exhaustion, depersonalization, reduced accomplishment, counter-lexicon).
- Implemented 3-layer PII anonymization (Presidio + spaCy NER + regex), GDPR-compliant data handling, and a 3-tier risk scoring system (0–100) capable of flagging burnout signals ≥ 3 weeks before self-reported crisis, with SHA-256 hash-chain audit logging across all predictions

Privacy-Preserving Health Monitor

Full Stack Developer

- Designed a camera-free elderly monitoring system with a CNN-LSTM fall detection model targeting $\geq 93\%$ sensitivity and $\leq 3\%$ false positive rate, ± 2 BPM heart rate accuracy via micro-Doppler phase extraction, and ≤ 60 -second end-to-end alert delivery latency.
- Delivered a full-stack system across 4 Dockerized microservices (FastAPI backend, React/TypeScript dashboard, FL aggregation server, edge runtime), exposing 14 REST API endpoints, 3 alert channels (Twilio SMS, Firebase push, local buzzer), and behavioral anomaly detection over a 2–4 week personal baseline learning window

SKILLS

Languages: Python, Java, SQL (PostgreSQL, MySQL), TypeScript, JavaScript (React.js, React Native), HTML/CSS.

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, OpenCV, FastF1, ETL Pipelines.

Tools: Docker, Git/GitHub/GitLab, CI/CD, Jupyter Notebook, VS Code, Postman.

CERTIFICATIONS

- Oracle Cloud Infrastructure Generative AI Professional** | Oracle
- Google Data Analytics Professional** | Google